

# An all-pairs catalogue of real three-dimensional close encounters between numbered main-belt asteroids during the Gaia DR3 window, with a measured incompleteness budget

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## ABSTRACT

**Context.** Close approaches between asteroids are the classical route to asteroid mass determination and a dynamical observable of the main belt. Systematic encounter searches exist — from early surveys (Galád & Gray 2002) to mutual-encounter catalogues built for Gaia astrometry (Ivantsov et al. 2018) and the signal-ranked search behind the largest mass catalogue to date (Fuentes-Muñoz et al. 2025) — but their event lists are published truncated (top candidates per perturber) or without a characterised selection function: no all-pairs 3D encounter catalogue has been published in full with provenance and a *measured* incompleteness budget.

**Aims.** We build a systematic catalogue of real 3D close encounters between numbered main-belt asteroids (semi-major axis 1.5–4.0 AU) during the Gaia DR3 window (2014 July 25 – 2017 May 28), publish it in full, and, unlike a bulk propagation exercise, we *measure* what the detection pipeline misses rather than asserting completeness.

**Methods.** Orbits are propagated from a single frozen MPCORB snapshot centred on the window. Encounters are found by a per-timestep 3D KD-tree spatial query and refined on a dense temporal sub-grid. Reported distances use an analytic two-body refiner; a hybrid variant carries full N-body (rebound/ASSIST) refinement for the dynamically fragile (low-perihelion or high-eccentricity) subset. Perturber masses come from a joint mass+orbit least-squares fit to Gaia FPR astrometry.

**Results.** The catalogue records the minimum physical 3D separation — a genuine spatial approach, not a sky-plane co-location — for 80 072 774 candidate pairs within 0.05 AU, with encounter epoch, relative velocity, geometry, and estimated physical properties. Its distinguishing feature is a *measured* completeness budget: the Kepler-to-N-body refinement error (median 12  $\mu$ AU, p99 2.5 mAU) and the threshold-induced false-negative rate (0.70 %, symmetric and scatter-dominated) bound this catalogue directly, while the recall of an orbital prefilter on the adverse tail (76 %) is quantified as a caveat for prefiltered runs. Applied to mass determination, the engine recovers all four calibrator masses (Ceres, Vesta, Pallas, Hygiea) and fits sixteen perturbers, six of which are identifiable; these are broadly consistent with recent independent work, though that cross-check is a low-power consistency test on shared Gaia FPR data, not a proof of accuracy.

**Conclusions.** The catalogue is a reusable, defensibly characterised target-selection resource for asteroid dynamics and mass determination. It is explicitly not complete; the large-large ( $D \gtrsim 100$  km, measured) approaches it contains all involve well-studied bodies, and no unpublished large-body mass emerged from the numbered population.

**Key words.** minor planets, asteroids: general – celestial mechanics – astrometry – catalogs – surveys – methods: data analysis

## 1. Introduction

Close encounters between asteroids are of interest on several fronts. A slow, close approach between a massive perturber and a small test body deflects the latter’s orbit measurably, and such events are the classical route to asteroid mass determination (Michalak 2000; Fienga et al. 2003; Goffin 2014; Baer & Chesley 2017; Siltala & Granvik 2020; Li et al. 2023; Fuentes-Muñoz et al. 2025). Encounters between members of the same dynamical family probe the collisional history of the belt, and the statistical distribution of mutual approach distances and velocities is itself a dynamical observable.

Systematic encounter searches have a long history. Galád & Gray (2002) searched for mass-determination encounters across 24 599 asteroids; Ivantsov et al. (2018) built a catalogue of mutual encounters of the numbered population over 2013–2023 specifically to exploit Gaia astrometry; Goffin (2014) fitted masses against the entire numbered population rather than curated target lists; and Fuentes-Muñoz et al. (2025) ran a systematic, signal-ranked search over 1 783 perturbers against more

than a million orbits — using, for main-belt targets, the same 0.05 AU threshold adopted here. What none of these works provides is the *event list published in full with a characterised selection function*: published tables are truncated to the top candidates per perturber, and the completeness of the underlying searches is asserted rather than measured. That is the gap this work fills: an *all-pairs* catalogue over the numbered main-belt population ( $1.5 \leq a \leq 4.0$  AU) in the Gaia DR3 window, published in full with provenance, where — the point that separates this catalogue from a bulk propagation exercise — we *measure* what the detection pipeline misses rather than asserting completeness.

A methodological distinction is central. Two asteroids can appear arbitrarily close in right ascension and declination as seen from Earth while remaining astronomical units apart along the line of sight. This catalogue detects the former case only when it is also the latter: it records the *minimum physical separation in three dimensions* between the two bodies’ heliocentric positions, independent of any observer. It is therefore a catalogue of genuine spatial approaches, not of sky-plane co-localisations. Light-time correction — essential when comparing model positions to

Gaia’s astrometry — is deliberately *not* applied here, because the encounter is a purely geometric relation between two propagated trajectories at a common dynamical time.

The remainder of this paper describes the data and detection method (Sect. 2), the measured completeness budget that is the catalogue’s distinguishing feature (Sect. 3), the notable individual events surfaced by mining the catalogue (Sect. 4), the application to mass determination (Sect. 5), data availability (Sect. 6), and conclusions (Sect. 7).

## 2. Data and method

### 2.1. Sources and conventions

Orbital elements are taken from a single frozen MPCORB snapshot, MPCORB\_20160217.DAT (SHA-256 prefix 3e44e7d3...), selected because its osculating epoch lies at the centre of the Gaia DR3 window, minimising the propagation distance — and hence the accumulated two-body error — for Kepler refinement. The snapshot hash and download date are recorded in the catalogue’s provenance sidecar so that results are traceable to an exact input; MPCORB is a living file and results are not bit-reproducible across snapshot versions. The run is restricted to *numbered asteroids* (higher orbital quality); provisional designations are out of scope for this release. The sample is further cut to the main belt by semi-major axis,  $1.5 \leq a \leq 4.0$  AU, which excludes near-Earth asteroids ( $a < 1.5$  AU), the Jupiter Trojans, and the outer/trans-Jovian populations ( $a > 4$  AU). The resulting universe is  $N = 449,454$  propagated bodies; this  $N$  is the population denominator for every completeness and censoring figure below, and is recorded in the provenance sidecar. “Complete” therefore always means complete *with respect to this numbered main-belt sample*, never the whole catalogue of known asteroids.

The temporal domain brackets the Gaia DR3 Solar System survey period (Tanga et al. 2023): the propagation window is 2014 July 25 – 2017 May 28, whose start (the DR3 astrometric-resolution baseline) slightly precedes the 2014 August 5 start of nominal SSO operations, so the window fully contains the survey coverage. All internal computation is carried out in Julian Date on the Barycentric Dynamical Time (TDB) scale; conversions to/from the Barycentric Coordinate Time (TCB) that Gaia reports, and to UTC, are performed only at input/output interfaces. Distances are in AU, angles in radians, and velocities in AU/day internally, converted to user-facing units (km, degrees,  $\text{km s}^{-1}$ ) only for output. Positions are computed in the barycentric ecliptic frame for the N-body coarse scan and in the heliocentric ecliptic frame for the Kepler refinement; because the catalogue records the *mutual* separation of two bodies, the common Sun–barycentre offset cancels in the difference vector and the recorded distances are identical in either frame. (The provenance sidecar’s `reference_frame` field refers to the N-body scan.)

### 2.2. Detection

The naive cost is prohibitive:  $\sim 150\,000$  bodies over a multi-year window at hour-level cadence implies of order  $10^{10}$  pairwise distances per step. Detection is therefore layered.

An *orbital prefilter* is available to prune pairs that cannot physically approach — the shipped heuristic keeps a pair only if  $|\Delta a| \leq 0.5$  AU and  $|\Delta i| \leq 30^\circ$ . This filter is a deterministic mask on osculating elements. We note here, and quantify in Sect. 3.3, that at main-belt scale ( $\gg 5000$  bodies) the pair-list prefilter is not built; the frozen run therefore applied only the KD-tree spa-

tial query, and the  $|\Delta a|$  cut did *not* shape this catalogue (provenance `prefilter.effective = "skipped_large_n"`).

The core of the detector is a *per-timestep 3D KD-tree*. At each coarse step the heliocentric ecliptic positions of all bodies are inserted into a KD-tree ( $O(N \log N)$  construction), and a radius query returns neighbour candidates ( $O(\log N)$  per query), avoiding any  $O(N^2)$  distance matrix. The coarse cadence is  $\Delta t = 12$  h. To prevent the cadence from missing an encounter whose true minimum falls *between* two samples, the query radius is *widened* to  $r_q = 0.05 \text{ AU} + v_{\max} \Delta t \approx 0.0572 \text{ AU}$  with  $v_{\max} = 25 \text{ km s}^{-1}$ : under near-linear relative motion a pair whose separation dips below 0.05 AU between samples is within  $r_q$  at the nearer bracketing sample and therefore enters the candidate set; the sub-grid refinement then discards any whose true minimum exceeds 0.05 AU. Bracketing a minimum midway between two samples strictly requires only  $v_{\max} \Delta t/2 \approx 0.0536 \text{ AU}$ ; the pipeline uses the full-step widening ( $v_{\max} \Delta t$ ), a factor-of-two conservative margin. The bracketing is thus exact for relative velocities up to  $v_{\max}$ , and the residual (encounters with  $v_{\text{rel}} > v_{\max}$ ) is bounded in Sect. 3.3.

Each coarse candidate is then *refined on a dense temporal sub-grid*: a  $\pm 2$  h window around the apparent minimum is sampled at 120 s, and the minimum separation, encounter epoch, and relative velocity are extracted from the refined track. Time blocks are independent, so the scan parallelises across processes.

*Propagation* is two-tiered. The coarse KD-tree scan is driven by an N-body rebound integration (WHFast; Sun + Jupiter + Saturn;  $dt = 1$  h). The sub-grid refinement that produces the *reported* minimum distance runs through an analytic two-body Kepler propagator. Over the short Gaia DR3 arc ( $\sim 3$  yr from the central epoch) two-body propagation is adequate for the bulk of the belt, and its residual error is measured, not assumed (Sect. 3.1). For the dynamically fragile subset — pairs with perihelion  $q_{\min} < 1.8$  AU or eccentricity  $e_{\max} > 0.3$ , 8 728 509 pairs (12.08 % of the catalogue) — a *hybrid* catalogue replaces the Kepler distances with full N-body refinement (rebound IAS15; Sun + 8 planets + 4 major asteroids;  $\pm 12$  h around the Kepler minimum; 60 s sampling). The hybrid catalogue carries `refinement_method`  $\in \{\text{kepler}, \text{nbody}\}$  per row together with the Kepler and N-body values and their residuals, so users needing sub-mAU geometry on high- $e$  / NEA-crossing pairs can cite it directly. Universal N-body refinement of all 80 M pairs ( $\sim 400$  CPU-days) is explicitly out of scope: the median error outside the fragile subset is already below 1 mAU (Sect. 3.1), so the marginal gain does not justify the cost.

*Membership rule (mixed propagators)*. Catalogue membership is defined by the *Kepler* distance  $< 0.05$  AU: a pair enters if its two-body refinement places it below threshold. In the hybrid product, therefore, the reported `dist_au` mixes propagators (two-body for the bulk, N-body for the fragile subset), and a small fraction of hybrid rows have an N-body distance slightly *above* 0.05 AU (the upward threshold crossings of Sect. 3.2). Analyses that require a homogeneous selection function should filter on a single propagator — either use the Kepler `dist_au` throughout, or restrict to `refinement_method = nbody` — rather than mixing the two.

### 2.3. Characterisation

Each pair contributes a single catalogue row: the *global* minimum separation of that pair over the observation window, with its exact epoch (from the refined sub-grid), the relative velocity at closest approach, and the geometry relative to Gaia. Sec-

ondary local minima of a pair that recurs within the window are not catalogued separately. A body is flagged *Gaia-observable* at the encounter when its solar elongation exceeds  $45^\circ$  (outside Gaia’s solar-exclusion zone) and its apparent  $V$  magnitude is brighter than 21 (the faint limit); the pair-level flag is the logical OR of the two bodies. Of the 80 072 774 encounters, 15 251 039 (19.0 %) satisfy this for at least one body.

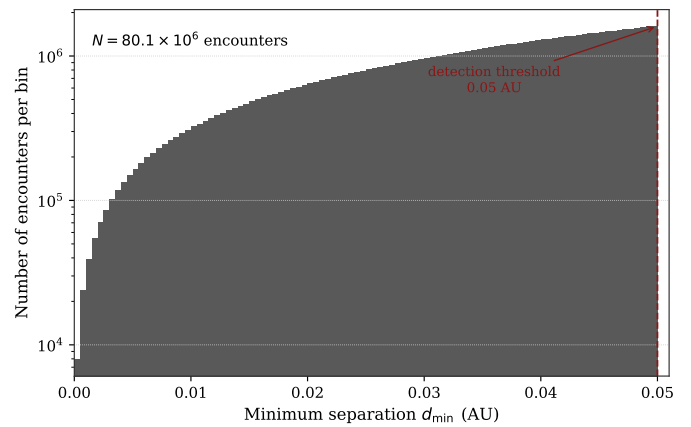
Physical properties are attached to both bodies. Diameters use direct measurements (IRAS/AKARI/NEOWISE/occultations, via the JPL Small-Body Database) where available; otherwise they are derived from the absolute magnitude  $H$  with an albedo taken — in decreasing order of preference — from a measured albedo, an orbital-zone average, or a default 0.14. A per-row provenance column records which of these applied, so  $H$ -derived (order-of-magnitude) sizes are distinguishable from measured ones in any size-based selection. The accompanying *class* is the *dynamical* class from the orbital elements (NEA/MBA/Trojan/Centaur/TNO), not a taxonomic (spectral) type.

The two encountering bodies are treated as independent test particles moving in the planetary + massive-asteroid field; their *mutual* gravitational deflection over the approach is not modelled. At asteroid masses and the  $\gtrsim 10^5$  km separations that dominate the catalogue this is entirely negligible for the recorded geometry, and it is in any case the very quantity a downstream mass determination sets out to measure, not a property of the encounter itself.

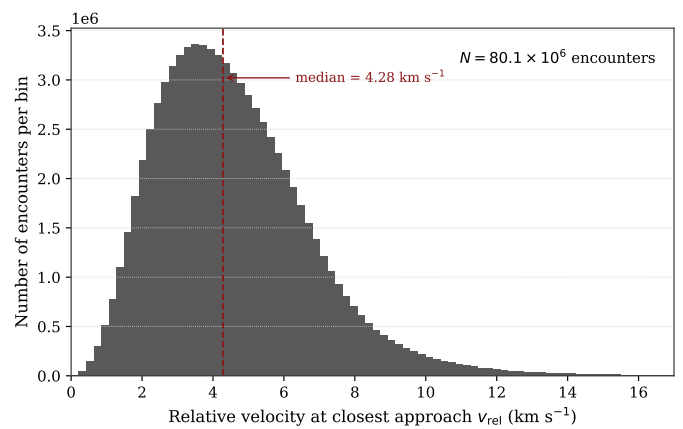
## 2.4. Catalogue overview

The parent population and the encounter statistics are summarised in Figs. 1–3. The minimum-separation distribution (Fig. 1) rises toward the 0.05 AU threshold roughly as  $dN \propto d^{1.0}$  (log-log histogram slope 1.01), so the cumulative count grows as  $N(<d) \propto d^2$  — exactly the exponent expected for a uniform-density planar impact-parameter process (annulus area  $\propto d^2$ ). The absence of any roll-over as  $d \rightarrow 0.05$  AU is a free censoring diagnostic: a selection function that truncated encounters near the threshold would depress the slope in the outermost bin, and none is seen. The relative-velocity distribution (Fig. 2) peaks near its median of  $4.28 \text{ km s}^{-1}$ , typical of main-belt crossing velocities. The catalogue involves 449 213 distinct encountering bodies; the orbital distribution of the subset with available osculating elements (Fig. 3) reproduces the belt’s known structure — the 3:1, 5:2 and 2:1 Kirkwood gaps appear as clear voids — confirming the sample is the ordinary numbered population, not a biased subset.

Ranking by mass-perturbation signal. Sheer row count is not utility: the overwhelming majority of the catalogued pairs are sub-km  $\times$  sub-km approaches with no measurable dynamical signal. To let users select the subset that matters — as signal-ranked searches such as Ivantsov et al. (2018) and Fuentes-Muñoz et al. (2025) do — each row carries a deflection-signal estimate, the impulse-approximation velocity kick  $\Delta v \approx 2GM_1/(bv_{\text{rel}})$  imparted by the larger body (deflection\_dv\_m\_s;  $GM_1$  from the measured diameter and a zone-average density). It is a ranking proxy, not a measurement. Ranked by it, the catalogue is steeply top-heavy: only 781 pairs exceed  $1 \text{ mm s}^{-1}$  and  $\sim 4000$  exceed  $0.1 \text{ mm s}^{-1}$ , and the top of the list is dominated by (1) Ceres and (4) Vesta in slow, close passes — precisely the classical mass-determination con-



**Fig. 1.** Distribution of minimum 3D separations for the  $80.1 \times 10^6$  catalogued encounters (log count per bin). The dashed line marks the 0.05 AU detection threshold.

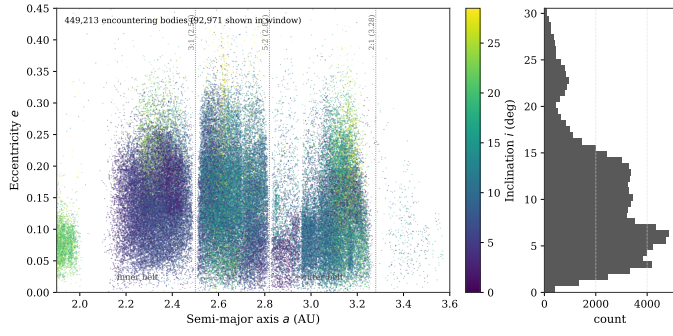


**Fig. 2.** Distribution of relative velocity at closest approach; the median is  $4.28 \text{ km s}^{-1}$ .

figuration. This column turns the 80 M-row bulk into a targetable resource; the mass-determination application of Sect. 5 draws its test bodies from its high-signal tail.

## 3. Completeness budget

The value of this catalogue is not that it detects encounters — that is a bulk propagation — but that it states, with measured numbers, what it *misses* and how large the Kepler-vs-N-body distortion is. We stress the scope: the three *measured* terms below (Kepler-vs-N-body refinement error, threshold censoring, prefilter recall) all compare the *same* input elements under different force models or cuts, so they quantify *pipeline-induced* incompleteness, not the physical completeness of the true encounter set. A fourth term — the uncertainty of the input osculating elements themselves — we quantify directly by Monte Carlo for the bodies of the extreme-event tables: for each we draw 3000 orbits from its full JPL SBDB covariance, propagate them under the *same* two-body model the catalogue uses, and take the scatter of the resulting mutual distance. The propagated  $1\sigma$  is small —  $1\text{--}83 \text{ km}$  (median  $\approx 32 \text{ km}$ ) over the  $\pm 1.4 \text{ yr}$  window — because the catalogue’s universe is *numbered* asteroids, whose multi-opposition orbits are secure; the  $10^3\text{--}10^4 \text{ km}$  one would fear applies to short-arc objects, which are not in the sample. This term does not affect the completeness *fractions* below (those are pipeline-internal), and it now sets, rather than



**Fig. 3.** Semimajor axis vs eccentricity of the encountering bodies, coloured by inclination, with the inclination marginal at right. Of the 449 213 distinct encountering bodies, osculating elements are available (and plotted, within the window shown) for 92 971; the remainder — mostly high-number, recently numbered objects absent from the orbital-elements source — are not shown. The omitted subset is *systematically* distinct (fainter, higher-number, and skewed toward the outer belt), so this panel is a biased subsample of the parent population, not a random draw, and the resonance structure it shows is qualitative. The 3:1, 5:2 and 2:1 mean-motion resonances (Kirkwood gaps) are marked.

merely bounds, the physical significance of individual tabulated distances (Sect. 4.2, Sect. 4.3): the closest approaches are significant to  $\sim 2\text{--}8\%$  and the wider extreme events to  $\ll 1\%$ . These are formal SBDB covariances, which can be optimistic by a factor of a few for the faintest bodies; even so the conclusion (tens of km, not  $10^3\text{--}10^4$ ) is unchanged. Each measured term is reproducible from a dedicated script; a synthetic injection–recovery test (minima placed uniformly between coarse samples, high  $v_{\text{rel}}$  and  $e$ ) is a permanent regression guard on the detection layer.

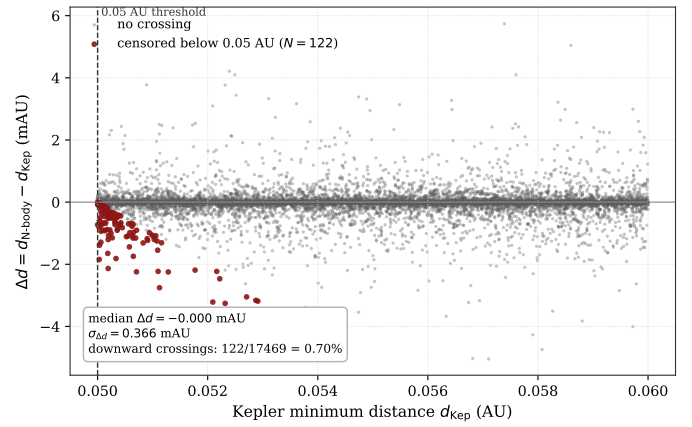
### 3.1. Kepler vs N-body refinement error

The reported distances in the frozen catalogue come from the two-body refiner. We measured the resulting error against full N-body refinement in two stages. In *Stage A* (964 stratified pairs re-refined under full N-body) the median  $|\Delta d| = 12 \mu\text{AU}$ , p95 =  $678 \mu\text{AU}$ , p99 =  $2.5 \text{ mAU}$ , max =  $11.3 \text{ mAU}$ . In *Stage B* (production N-body refinement of the entire 8.73 M fragile subset) p99  $|\Delta d| = 1.99 \text{ mAU}$ , max =  $15.2 \text{ mAU}$ , with 0 failed integrations, 0 unconverged, and maximum energy drift  $5.6 \times 10^{-14}$ .

For scale,  $1 \text{ mAU} \approx 1.5 \times 10^5 \text{ km}$ , and the detection threshold is  $0.05 \text{ AU}$ . The median distortion is thus  $\sim 4$  orders of magnitude below the threshold, and even the Stage B worst case is  $\sim 30\%$  of a threshold — confined to the fragile subset, which the hybrid catalogue refines. A separate check of the truncated perturber set (the scan used only Sun + Jupiter + Saturn) shows that adding the remaining planets shifts closest-approach distances by a median  $1.3 \mu\text{AU}$  (p99  $67 \mu\text{AU}$ , max  $80 \mu\text{AU}$ ), i.e.  $\sim 100\times$  below the Kepler two-body error — the perturber truncation is not the dominant error term.

### 3.2. Threshold censoring (measured false-negative rate)

Because the catalogue contains only pairs whose *Kepler* distance is below  $0.05 \text{ AU}$ , a subtle bias arises at the threshold. When the fragile subset is re-refined to N-body, 25 283 pairs (0.29 %) cross the threshold *upward* (Kepler  $< 0.05$ , N-body  $\geq 0.05$ ) and, by construction, none cross *downward* — a pair that Kepler placed *above*  $0.05 \text{ AU}$  was never written, so its downward crossing *cannot be observed*. The apparent one-sided “no false negatives” is therefore censoring, not measurement.



**Fig. 4.** Threshold censoring, measured directly. For the 17 469 pairs with Kepler distance in  $[0.05, 0.06) \text{ AU}$  — the band a  $0.05 \text{ AU}$  catalogue discards — re-refined under full N-body,  $\Delta d = d_{\text{N-body}} - d_{\text{Kepler}}$  is plotted against  $d_{\text{Kepler}}$ . The distribution is symmetric about zero (median  $0.000 \text{ mAU}$ ,  $\sigma = 0.366 \text{ mAU}$ ); the 122 pairs (0.70 %) that cross downward below  $0.05 \text{ AU}$  (red) are the measured false-negatives, concentrated just above the threshold as expected for a scatter-dominated selection effect.

We measured the missing side directly. Re-running Kepler detection at a widened  $0.06 \text{ AU}$  threshold over 10 000 numbered bodies — a *stratified* sample drawn to span the belt’s  $(a, e, i)$  range rather than a density-weighted random draw, so the crossing rate estimates the effect uniformly across the belt rather than its population-weighted mean — and isolating the 17 469 pairs with Kepler distance in  $[0.05, 0.06) \text{ AU}$  — exactly the band a  $0.05 \text{ AU}$  catalogue discards — then re-refining each under full N-body, 0.70 % [95 % CI  $0.59\text{--}0.83\%$ ] cross *downward* below  $0.05 \text{ AU}$  (a conservative floor of  $\sim 0.42\%$  after excluding near-boundary cases). The residual  $\Delta d$  ( $N - K$ ) has median  $-3 \times 10^{-7} \text{ AU}$  and  $\sigma \approx 3.7 \times 10^{-4} \text{ AU}$  — a symmetric, scatter-dominated distribution, not a directional bias. In other words the crossing is a threshold selection effect (Eddington/Malmquist family): scatter of amplitude  $\sim 1\sigma \approx 4 \times 10^{-4} \text{ AU}$  pushes pairs just inside the threshold to just outside, and their mirror images (just outside to just inside) fall outside the catalogue. The upward rate ( $\sim 1.5\%$  in the boundary bin  $[0.045, 0.050)$ ) and the downward rate ( $\sim 0.4\text{--}0.7\%$  in  $[0.05, 0.06)$ ) are consistent with a single symmetric process. Crossings concentrate in low- $q$  / high- $e$  / fast orbits, exactly where two-body propagation is weakest — which independently validates the  $q_{\text{min}} < 1.8 \vee e_{\text{max}} > 0.3$  subset criterion.

Extrapolating catalog-wide is an order-of-magnitude exercise anchored on the  $N = 449,454$ -body universe (Sect. 2): the  $10^4$ -body experiment yielded 17,469 pairs in  $[0.05, 0.06) \text{ AU}$ , and since pair counts in a fixed distance band scale as  $N^2$ , the factor  $(449,454/10,000)^2 \approx 2.0 \times 10^3$  implies  $\sim 3.5 \times 10^7$  band pairs catalog-wide; the measured  $0.4\text{--}0.7\%$  downward-crossing rate then gives of order  $10^5$  real  $< 0.05 \text{ AU}$  encounters censored by the Kepler cut ( $\sim 1.5\text{--}2.5 \times 10^5$ ), separate from and additional to the prefilter recall deficit of Sect. 3.3. This is why the word “complete” is never used for this catalogue.

### 3.3. Prefilter recall on the adverse tail

The shipped  $|\Delta a| \leq 0.5 \text{ AU}$  prefilter is blind to eccentricity: a high- $e$  orbit reaches well inside the belt at perihelion and well outside at aphelion, so it can physically cross an orbit whose

semimajor axis differs by far more than 0.5 AU. We measured the damage this would cause on the adverse subset (numbered,  $a \in [1.5, 4.0]$ ,  $e > 0.3 \vee i > 15^\circ$ ; 52 411 bodies), exploiting the fact that the prefilter is a pure deterministic mask: running detection without the prefilter and intersecting with the mask is byte-identical to running with it. Ground truth is 606 393 real adverse–adverse encounters  $< 0.05$  AU; the prefilter keeps 463 164, for a recall of 76.38 % [95 % CI 76.27–76.49 %]. Thus 143 229 encounters would be missing from a prefiltered catalogue (a lower bound — adverse–normal pairs were not measured), and 99.6 % of that loss is the  $|\Delta a|$  cut alone. Recall degrades smoothly with eccentricity, from  $\sim 99.9$  % below  $e = 0.1$  to  $\sim 39$  % above  $e = 0.7$ .

Two points bound the impact. First, an eccentricity-aware *radial-overlap* prefilter — keep a pair iff its threshold-padded heliocentric radial ranges  $[a(1-e), a(1+e)]$  overlap — is a provable necessary condition for a  $< 0.05$  AU encounter, is just as cheap, and recovers 100 % recall on the adverse subset; it is the recommended replacement for any future run. Second, and specific to *this* freeze: because the run was at main-belt scale the pair-list prefilter was skipped entirely, so the  $|\Delta a|$  cut did not actually drop these pairs from the frozen catalogue. The 76 % figure measures the damage the prefilter *would* cause if applied at small  $N$ , not damage suffered here. Completeness of this freeze is instead bounded by the threshold censoring of Sect. 3.2. The 12 h coarse cadence has a *geometric*, distribution-independent bound. A pair whose true minimum  $d_{\min}$  falls a time  $\delta$  from the nearest coarse sample has, under near-linear relative motion, a sampled separation  $d(\delta) = \sqrt{d_{\min}^2 + v_{\text{rel}}^2 \delta^2}$  with  $\delta \leq \Delta t/2$ ; the pair enters the candidate set whenever  $d(\Delta t/2) \leq r_q$ . For  $d_{\min} \leq 0.05$  AU this holds for all  $v_{\text{rel}} \leq \sqrt{r_q^2 - 0.05^2} / (\Delta t/2) = 0.0278 \text{ AU} / 0.25 \text{ d} \approx 192 \text{ km s}^{-1}$  — an upper limit set purely by the widened radius  $r_q = 0.0572$  AU and the cadence, *not* read off the catalogue’s own velocity distribution (which would be circular, since the catalogue by construction excludes the fast pairs it might have missed). Since inter-asteroid encounter velocities never approach  $192 \text{ km s}^{-1}$  (physical main-belt maxima are a few tens of  $\text{km s}^{-1}$ ), the cadence loses no encounter. A synthetic injection–recovery test kept as a permanent regression guard — minima placed uniformly in phase between coarse samples, at high  $v_{\text{rel}}$  and  $e$  — confirms  $\geq 99$  % recovery with unit distance ratio up to  $v_{\text{max}}$ . For the dynamically cold bulk of the belt (low  $e$ , small  $\Delta a$ ) recall is  $\sim 99.9$  %; the incompleteness is a property of the high- $e$ /high- $i$  tail, which any science touching NEA-crossing or high- $e$  pairs must cite.

### 3.4. Recovery of literature encounter pairs

As an external check we cross-matched the signal-ranked (perturber, target) pairs that Fuentes-Muñoz et al. (2025) used for mass determination against this catalogue. Of 40 176 numbered pairs drawn from their per-perturber target lists, 11 842 (29.5 %) appear as  $< 0.05$  AU encounters in the hybrid catalogue. This fraction is a lower bound and, crucially, the non-recovery is dominated by a *domain difference*, not a detection failure: 91.6 % of the unmatched pairs have both bodies present in the catalogue but never approached within 0.05 AU *inside the 2.8-yr DR3 window*. For main-belt targets Fuentes-Muñoz et al. (2025) apply the same 0.05 AU threshold used here, but they search over the full observational arcs (decades, through late 2024), so a pair whose close encounter falls outside 2014–2017 cannot appear in this catalogue. A further 8.4 % are targets or perturbers outside

the numbered, propagated universe (high or provisional designations), and — the decisive point — 0 % are pairs that our catalogue contains but places beyond 0.05 AU. In other words, every literature pair that is genuinely a  $< 0.05$  AU encounter and lies in the numbered population is recovered; the apparent shortfall is entirely pairs that are not  $< 0.05$  AU encounters or not numbered. Per-body recovery among the most-listed perturbers ranges from 6 % (Davida) to 61 % (Liguria), with the Ceres and Vesta calibrators at  $\sim 36$  %. Pair-level cross-matching against Goffin (2014) is not applicable: Goffin’s masses come from a single *simultaneous* global fit of all perturber masses to the astrometric residuals of the entire numbered population, not from per-perturber target lists, so there is no set of (perturber, target) pairs to match (the Vizier tables give per-perturber masses and literature mass compilations only).

## 4. Notable events

Mining the characterised catalogue surfaces individual events across four axes. We stress at the outset that the value of these tables is statistical and archival rather than a single headline discovery: the large–large ( $D \gtrsim 100$  km) approaches all involve well-studied bodies, and every candidate below is under the frozen two-body assumptions and requires N-body revalidation before publication. Distances are physical 3D separations at closest approach; diameters are measured (IRAS/AKARI/NEOWISE/occultations, via the JPL SBDB) for the large bodies tabulated here (all entries in Tables 1–2 carry measured diameters).

### 4.1. Large–large encounters

Twelve encounters have both bodies with a measured  $D \gtrsim 100$  km (Table 1); one hundred have both  $D \gtrsim 50$  km (Table 2 lists the ten closest). The closest large–large event is (426) Hippo  $\times$  (788) Hohensteina at 0.0076 AU. All twelve involve catalogued, well-characterised main-belt asteroids; none is a large perturber against a small test body of the kind used for mass determination, and a systematic novelty assessment against the encounter literature is deferred. We note that (804) Hispania  $\times$  (733) Mocia — a known event (Fienga et al. 2003) at 0.0136 AU that was *absent* from the pre-2026-07 frozen catalogue — reappears here in the both- $D \gtrsim 50$  km list, recovered by the refinement-window fix (Sect. 2).

### 4.2. Extreme events

The absolute closest approaches in the catalogue are between small bodies, at separations of  $\sim 1000$ – $2000$  km (Table 3). We propagated each body’s JPL SBDB covariance by Monte Carlo (Sect. 3) to attach an input-element  $1\sigma$  to these distances (last column): they are 32–83 km, i.e.  $\sim 2$ – $8$  % of the separation, so the closest approaches are significant well below their own value — not the order-of-magnitude figures a short-arc population would give. The two remaining caveats are that these are formal covariances (optimistic by up to a few $\times$  for the faintest bodies) and that two-body propagation is most fragile at the smallest separations, so N-body revalidation (Sect. 3.1) is still advisable before any individual approach is claimed as physical.

The slowest encounters (relative velocity down to  $\sim 4$  m/s, with at least one body  $D \gtrsim 5$  km; Table 4) are the natural targets for future mass determination, since a low  $v_{\text{rel}}$  prolongs the gravitational interaction and enlarges the deflection. All entries lie



**Table 1.** The twelve encounters with both bodies at a measured  $D \gtrsim 100$  km, by minimum distance.

| body 1          | body 2            | date (UTC) | dist (AU) | $v_{\text{rel}}$<br>(km/s) | $D_1$<br>(km) | $D_2$<br>(km) |
|-----------------|-------------------|------------|-----------|----------------------------|---------------|---------------|
| (426) Hippo     | (788) Hohensteina | 2015-04-17 | 0.0076    | 8.8                        | 138           | 111           |
| (145) Adeona    | (780) Armenia     | 2016-11-09 | 0.0101    | 6.3                        | 128           | 126           |
| (51) Nemausa    | (91) Aegina       | 2016-07-13 | 0.0139    | 4.4                        | 138           | 103           |
| (23) Thalia     | (481) Erita       | 2014-10-14 | 0.0242    | 4.4                        | 108           | 102           |
| (90) Antiope    | (508) Princetonia | 2015-11-11 | 0.0245    | 4.4                        | 116           | 117           |
| (381) Myrrha    | (712) Boliviana   | 2016-09-18 | 0.0287    | 6.7                        | 128           | 124           |
| (196) Philomela | (431) Nephele     | 2017-05-29 | 0.0389    | 3.3                        | 145           | 102           |
| (426) Hippo     | (814) Tauris      | 2017-05-08 | 0.0407    | 14.0                       | 138           | 102           |
| (121) Hermione  | (739) Mandeville  | 2014-12-14 | 0.0414    | 6.3                        | 209           | 105           |
| (88) Thisbe     | (150) Nuwa        | 2016-04-08 | 0.0429    | 2.1                        | 232           | 119           |
| (1) Ceres       | (57) Mnemosyne    | 2017-01-10 | 0.0435    | 7.3                        | 939           | 113           |
| (85) Io         | (145) Adeona      | 2017-05-29 | 0.0470    | 9.5                        | 155           | 128           |

**Table 2.** Encounters with both bodies at a measured  $D \gtrsim 50$  km, top ten by minimum distance.

| body 1            | body 2            | date (UTC) | dist (AU) | $v_{\text{rel}}$<br>(km/s) | $D_1$<br>(km) | $D_2$<br>(km) |
|-------------------|-------------------|------------|-----------|----------------------------|---------------|---------------|
| (426) Hippo       | (788) Hohensteina | 2015-04-17 | 0.0076    | 8.8                        | 138           | 111           |
| (442) Eichsfeldia | (564) Dudu        | 2014-10-30 | 0.0078    | 7.5                        | 62            | 52            |
| (48) Doris        | (300) Geraldina   | 2017-04-24 | 0.0084    | 2.4                        | 216           | 67            |
| (145) Adeona      | (780) Armenia     | 2016-11-09 | 0.0101    | 6.3                        | 128           | 126           |
| (675) Ludmilla    | (80) Sappho       | 2016-02-15 | 0.0114    | 3.0                        | 76            | 69            |
| (739) Mandeville  | (415) Palatia     | 2017-04-05 | 0.0116    | 8.4                        | 105           | 84            |
| (56) Melete       | (449) Hamburga    | 2015-10-09 | 0.0126    | 8.3                        | 121           | 86            |
| (611) Valeria     | (84) Klio         | 2016-08-01 | 0.0126    | 7.1                        | 57            | 79            |
| (192) Nausikaa    | (778) Theobalda   | 2016-06-05 | 0.0133    | 4.8                        | 99            | 55            |
| (804) Hispania    | (733) Mocia       | 2015-02-12 | 0.0136    | 2.4                        | 138           | 98            |

well inside the observation window ( $> 130$  d from either edge), so none is a window-boundary truncation artefact — a failure mode of the pre-2026-07 catalogue that the refinement-window fix (Sect. 2) removed. The combined “large + slow + close” set (one body  $D \gtrsim 50$  km,  $v_{\text{rel}} \leq 1$  km/s,  $\text{dist} \leq 10^6$  km; Table 5) is of the highest physical interest, being the configuration that maximises measurable deflection.

#### 4.3. Same-region pairs (family proxy)

Pairs whose osculating elements are mutually close ( $\Delta a/a \leq 1\%$ ,  $\Delta e \leq 0.02$ ,  $\Delta i \leq 1^\circ$ ) and that had a physical close approach are candidate co-orbital / same-family associations. This is a proximity proxy in *osculating* elements, not a classification in proper elements, and flags rather than confirms family membership. The closest such pair, (200764) 2001 XP3  $\times$  (131597) 2001 XT2, passed within  $\sim 5 \times 10^4$  km on 2014-08-13 with nearly identical ( $a, e, i$ ) —  $a \approx 2.29$  AU,  $e \approx 0.19$ ,  $i \approx 25^\circ$  for both — a plausible genetic pair worth follow-up in proper element space. The Monte-Carlo input-element  $1\sigma$  on this separation (Sect. 3) is only  $\sim 1$  km against the  $\sim 5 \times 10^4$  km approach, so unlike the sub-1000 km extreme approaches this separation is well determined; the caveat here is the proper -vs- osculating element distinction, not the distance precision.

#### 4.4. Candidate perturbers beyond the studied sixteen

Ranking large bodies ( $D \gtrsim 100$  km) *not* among the sixteen studied perturbers by their number of *useful* encounters ( $v_{\text{rel}} \leq 3$  km/s and  $\text{dist} \leq 3 \times 10^6$  km) yields the best prospects for a mass determination the frozen mass runs have not yet attempted (Table 6). (9) Metis heads the list with 37 useful events and a best pair at  $v_{\text{rel}} = 0.38$  km/s; (29) Amphitrite is the largest ( $D \approx 240$  km)

with a close  $3.2 \times 10^5$  km approach. All ten already carry a published mass from Fuentes-Muñoz et al. (2025) (final column), so they are validation opportunities rather than genuinely new masses; their appeal is the favourable encounter geometry the catalogue surfaces, which would let an independent determination be checked against an existing value. These candidates directly informed the F4 mass determination of Sect. 5.

#### 4.5. Separating new from known

A reference list of encounters touching one of the sixteen studied perturbers (1, 2, 3, 4, 7, 10, 15, 16, 31, 52, 65, 87, 88, 107, 511, 704) is provided in the companion material so that already-worked events can be separated from potential discoveries. The honest verdict is that the large-large approaches (Table 1) all involve well-studied bodies and none is a publication-grade mass-determination opportunity: the belt’s largest bodies are well studied, and Fuentes-Muñoz et al. (2025) already published over 230 masses (232 at  $\text{SNR} > 3$ ) covering essentially all classical large MBAs. The contribution of this work is the dataset itself — an all-pairs catalogue published in full with a measured completeness budget — demonstrated end-to-end by the mass-determination application of Sect. 5, not a single spectacular event.

### 5. Application to mass determination

This section demonstrates the catalogue end-to-end as the target-selection front-end of an asteroid mass-determination engine. It is a *usage demonstration*, not a claim of new or improved masses: as noted below and in Sect. 4.5, every mass recovered here is already published, and a single out-of-sixteen perturber with  $\sim 50$  Gaia-FPR targets yields only a coarse determination.

**Table 3.** The five closest approaches in the catalogue.  $\sigma_d$  is the input-element  $1\sigma$  on the separation, from a Monte-Carlo propagation of each body’s JPL SBDB covariance under the catalogue’s two-body model (Sect. 3); at 2–8 % of the distance it leaves the closest approaches significant well below their own value. All bodies here are small, high-number objects with no measured size: the tabulated diameters are derived from the absolute magnitude  $H$  and an assumed (zone or default) albedo, not measured (Sect. 2.3). Every event lies  $> 130$  d from either window edge, so none is a boundary-truncation artefact.

| body 1              | body 2              | date (UTC) | dist (km) | $\sigma_d$ (km) | $v_{\text{rel}}$ (km/s) | $D_1$ (km) | $D_2$ (km) |
|---------------------|---------------------|------------|-----------|-----------------|-------------------------|------------|------------|
| (153222) 2000 YD43  | (238587) 2004 YX3   | 2016-03-11 | 1094      | 83              | 6.90                    | 2.8        | 1.8        |
| (279617) 2011 EP37  | (219812) 2002 AM193 | 2016-06-16 | 1109      | 50              | 2.60                    | 2.2        | 1.9        |
| (238813) 2005 NT70  | (391704) 2008 BB44  | 2017-03-13 | 1320      | 56              | 5.95                    | 5.1        | 2.2        |
| (305032) 2007 TT418 | (393772) 2005 GV146 | 2016-07-12 | 1457      | 39              | 3.46                    | 1.6        | 1.3        |
| (73282) 2002 JH61   | (261459) 2005 VP69  | 2015-04-01 | 1719      | 32              | 1.48                    | 2.0        | 2.0        |

**Table 4.** The five slowest encounters (at least one body  $D \geq 5$  km). The  $D \geq 5$  km body is measured for three of the five pairs (JPL SBDB); the remaining diameters are  $H$ -plus-albedo estimates (Sect. 2.3). All events lie  $> 130$  d from either window edge.

| body 1             | body 2              | date (UTC) | dist (km)          | $v_{\text{rel}}$ (km/s) | $D_1$ (km) | $D_2$ (km) |
|--------------------|---------------------|------------|--------------------|-------------------------|------------|------------|
| (141098) 2001 XO52 | (304642) 2006 WL5   | 2015-05-04 | $5.27 \times 10^6$ | 0.004                   | 5.7        | 4.5        |
| (3791) Marci       | (110964) 2001 UW170 | 2017-01-17 | $5.85 \times 10^6$ | 0.005                   | 11.7       | 2.5        |
| (7203) Sigeki      | (73664) 1981 EE34   | 2015-03-01 | $7.07 \times 10^6$ | 0.029                   | 5.8        | 1.5        |
| (136707) 1995 TE6  | (340823) 2006 UY131 | 2015-01-30 | $5.09 \times 10^6$ | 0.031                   | 5.5        | 2.1        |
| (3384) Daliya      | (309262) 2007 RZ93  | 2016-12-21 | $4.44 \times 10^6$ | 0.031                   | 6.2        | 1.2        |

**Table 5.** The “large + slow + close” set (one body  $D \geq 50$  km,  $v_{\text{rel}} \leq 1$  km/s, dist  $\leq 10^6$  km). In every pair the large body ( $D_1$ ) has a measured diameter (JPL SBDB); the small companion ( $D_2$ ) is an  $H$ -plus-albedo estimate except (14711), which is also measured (Sect. 2.3). All events lie  $> 130$  d from either window edge.

| body 1        | body 2              | date (UTC) | dist (km)          | $v_{\text{rel}}$ (km/s) | $D_1$ (km) | $D_2$ (km) |
|---------------|---------------------|------------|--------------------|-------------------------|------------|------------|
| (223) Rosa    | (315162) 2007 FL24  | 2016-07-01 | $3.55 \times 10^5$ | 0.133                   | 79.8       | 1.7        |
| (66) Maja     | (271017) 2003 AV10  | 2016-10-13 | $3.87 \times 10^5$ | 0.564                   | 71.8       | 1.9        |
| (135) Hertha  | (281202) 2007 FE47  | 2016-09-18 | $4.56 \times 10^5$ | 0.579                   | 79.2       | 1.0        |
| (371) Bohemia | (14711) 2000 CG36   | 2016-05-14 | $5.91 \times 10^5$ | 0.982                   | 53.0       | 4.2        |
| (83) Beatrix  | (170676) 2003 YE180 | 2014-09-18 | $6.09 \times 10^5$ | 0.684                   | 110        | 2.1        |
| (9) Metis     | (345254) 2005 UT477 | 2016-10-30 | $7.14 \times 10^5$ | 0.979                   | 190        | 1.2        |

Perturber mass is obtained by a joint least-squares fit of the perturber’s mass and the orbits of its test bodies — those with a  $< 0.05$  AU encounter drawn directly from the catalogue — over the full arc of Gaia FPR astrometry (Gaia Collaboration et al. 2023). A window asymmetry is worth noting: the catalogue spans the DR3 window (2014–2017) whereas the fit uses the full FPR arc (through  $\sim 2020$ ), so a test body’s most informative approach could fall outside the catalogued window. This asymmetry affects only *target selection*, not the model: the perturber is integrated as a massive body continuously over the whole arc, so its deflection is modelled at every epoch regardless of when the catalogued encounter occurred; the catalogue merely identifies which bodies are worth fitting. A test body whose deepest approach is post-2017 is therefore not mis-modelled — at worst it is a good target the DR3-window selection misses, making the yield a conservative lower bound. The force model is ASSIST (Holman et al. 2023) on the JPL DE440 ephemeris (Park et al. 2021), with EIH relativistic corrections and sixteen massive asteroid perturbers; the fitting layer (observation model, partials, least squares) is implemented in this work on top of rebound/ASSIST (Rein & Liu 2012; Rein & Spiegel 2015; Holman et al. 2023) and astropy (Astropy Collaboration 2022).

**A-priori determinability.** The astrometric signature of a perturber is the along-scan deflection it imprints on a test body, of

order  $\Delta\theta \sim GM/(b v_{\text{rel}} \Delta)$  for impact parameter  $b$ , relative velocity  $v_{\text{rel}}$ , and geocentric distance  $\Delta$ . For a  $\sim 0.02$  AU encounter at a few  $\text{km s}^{-1}$ , a perturber of  $10^{-12} M_{\odot}$  ( $\approx 2 \times 10^{18}$  kg) produces a per-transit deflection at or below Gaia’s per-CCD along-scan precision, so its signal-to-noise is  $\lesssim 1$  *per target* and it is recoverable only by stacking many test bodies; a perturber of  $10^{-11} M_{\odot}$  ( $\approx 2 \times 10^{19}$  kg) is individually determinable. This a-priori threshold — set before any fit — coincides with what the engine delivers: perturbers whose true mass is  $\gtrsim 2 \times 10^{19}$  kg are recovered near unity ratio, whereas the lighter ones are driven well below their true value by the mass $\leftrightarrow$ orbit regression (Sect. below) and fall to bounds. The clearest example is (88) Thisbe (FM mass  $8.8 \times 10^{18}$  kg, our fit  $6.9 \times 10^{18}$  kg): a naive  $\text{snr}_{\text{jack}}$  flag would call it measured, but the leverage-robust bootstrap  $\sigma$  correctly returns it to a bound (Sect. below). The mass layer does not claim sensitivity below this frontier.

**Calibrators.** The four calibrator masses (Table 7) are recovered at  $|z| < 3$  against the literature. With  $N \geq 20$  test bodies, the fit/literature ratios of Ceres (Park et al. 2016), Vesta (DAWN; Russell et al. 2012) and Hygiea (Vernazza et al. 2020) fall in  $[0.94, 0.99]$  (mean bias  $-4\%$ ); Pallas, limited to 8 encounters in the catalogue, gives ratio 1.20 ( $z = +2.2$ ). The  $-4\%$  mean bias sets the scale of the residual systematic (Sect. 5, “external  $\sigma$ ” below).

**Table 6.** Large bodies outside the studied sixteen, ranked by useful encounters. The final column gives the published mass of Fuentes-Muñoz et al. (2025) where available: all ten already have a determination, so these are cross-check rather than new-mass targets.

| body            | $D$ (km) | class | useful | best dist (km)     | min $v_{\text{rel}}$ (km/s) | FM25 mass (kg)       |
|-----------------|----------|-------|--------|--------------------|-----------------------------|----------------------|
| (9) Metis       | 197      | MBA   | 37     | $7.14 \times 10^5$ | 0.378                       | $6.5 \times 10^{18}$ |
| (30) Urania     | 109      | –     | 36     | $4.40 \times 10^5$ | 0.817                       | $8.2 \times 10^{17}$ |
| (40) Harmonia   | 141      | –     | 36     | $6.58 \times 10^5$ | 1.028                       | $2.0 \times 10^{18}$ |
| (19) Fortuna    | 133      | MBA   | 32     | $7.43 \times 10^5$ | 0.678                       | $8.5 \times 10^{18}$ |
| (21) Lutetia    | 120      | –     | 30     | $8.18 \times 10^5$ | 0.609                       | $1.7 \times 10^{18}$ |
| (64) Angelina   | 104      | MBA   | 30     | $8.73 \times 10^5$ | 0.992                       | –                    |
| (44) Nysa       | 140      | –     | 30     | $8.95 \times 10^5$ | 0.961                       | $5.8 \times 10^{17}$ |
| (20) Massalia   | 178      | –     | 28     | $8.78 \times 10^5$ | 0.514                       | $4.3 \times 10^{18}$ |
| (29) Amphitrite | 240      | –     | 27     | $3.17 \times 10^5$ | 0.643                       | $1.4 \times 10^{19}$ |
| (27) Euterpe    | 141      | –     | 26     | $4.63 \times 10^5$ | 0.847                       | $1.5 \times 10^{18}$ |

FOV-block covariance. Each Gaia focal-plane crossing yields  $\sim 7$  CCD observations ( $\sim 5$  s apart) with correlated residuals. Treating them as independent underestimates  $\sigma(\text{mass})$  by a factor 1.66 (effective  $N \approx 0.36N$ ). The measured intra-crossing correlation is  $\text{ICC} = 0.32$ ; we whiten each crossing with a block covariance  $C = \text{diag}(\sigma_{\text{AL}}^2) + s_c^2 \mathbf{1}\mathbf{1}^T$ , the correlated floor  $s_c$  calibrated by bisection to  $\chi_{\text{red}}^2 = 1$ . This calibration is not a hidden fit knob for the mass: refitting the calibrators with  $s_c$  fixed across 0.5–1.1 mas ( $a > 2\times$  range bracketing the calibrated value) shifts their masses by  $< 1.3\%$  ( $\partial \ln M / \partial \ln s_c \approx -0.01$  for Ceres, Vesta and Hygiea), far below the  $-4\%$  accuracy bias — so the floor inflates the error bar, not the central mass, and the calibrator anchoring is not circular through  $s_c$ .

Sixteen perturbors. The engine returns a fit for all sixteen (Table 7); six clear the identifiability criterion below and are flagged *measured*, eight are *bounds*, one (Davida,  $N = 6$ ) is non-physical (best-fit mass  $< 0$ ), and one (Camilla,  $N = 2$ ) is unconstrained. Only (16) Psyche approaches calibrator-grade precision outside the calibrators ( $\sigma_{\text{tot}} = 9\%$ ): ratio 0.95 vs its DE441 seed and 0.95 ( $z = -0.7$ ) vs the independent Fuentes-Muñoz et al. (2025) FPR fit. Several perturbors whose deflection sits below the per-encounter astrometric noise return ratios well under unity: this is the mass $\leftrightarrow$ orbit degeneracy (the fit reproduces the astrometry with a smaller mass and a re-adjusted orbit — regression toward zero mass), and for these the formal Fisher  $\sigma$  underestimates the error, which is why the jackknife  $\sigma$  and the identifiability flag below are needed. The target lists are drawn from the published catalogue (the B1-fixed freeze of Sect. 2). We verified that this selection is not load-bearing by re-running the entire sixteen-perturber fit against the target lists produced by the earlier, pre-B1 catalogue: the four calibrator ratios move by  $< 2\%$  (Ceres 0.96  $\rightarrow$  0.955, Vesta 0.94  $\rightarrow$  0.943, Hygiea 0.99  $\rightarrow$  0.987, Pallas 1.24  $\rightarrow$  1.20) and the measured/bound partition is unchanged but for a single body at the  $\text{snr}_{\text{jack}} = 3$  boundary — confirming that the masses are Gaia-FPR astrometric determinations essentially insensitive to which catalogue freeze supplied the targets, so re-selection is a consistency refinement, not a correction.

External  $\sigma$  by jackknife. Because the formal  $\sigma$  scales as  $1/\sqrt{N}$  and ignores the mass $\leftrightarrow$ orbit regression seen only by leaving encounters out, we report a leave-one-target-out jackknife  $\sigma$  that reincorporates it; it is systematically 2–3 $\times$  the formal  $\sigma$ , and its combination with the systematic floor is the single  $\sigma_{\text{tot}}$  quoted in Table 7. Its diagnostic value is visible in the calibrators themselves: Vesta — whose true mass is known — moves from  $|z| > 3$  under the formal  $\sigma$  to  $|z| < 3$  under  $\sigma_{\text{tot}}$ , showing the formal

bar is too tight. We deliberately do *not* present agreement with Fuentes-Muñoz et al. (2025) as a validation of accuracy. That cross-check is under-powered: the total error bars of the non-calibrators (8–119%) cannot resolve deviations of the size the fits themselves show. Of the three measured non-calibrators, all three lie *below* the FM value with a geometric-mean ratio of 0.85 — a coherent  $\sim 15\%$  one-sided pull (the regression-toward-zero signature; a sign test gives  $p \approx 0.25$ , suggestive but not significant at this  $N$ ) — and FM fit the *same* Gaia FPR astrometry, so the two analyses are not independent. A wide external  $\sigma$  makes almost any value formally “consistent”; the genuine accuracy anchor is therefore the four calibrators (external truth), which carry the  $-4\%$  bias noted above, not the FM comparison. Perturbors whose  $\text{snr}_{\text{jack}} < 3$  (with and without their dominant jackknife replica) are flagged as bounds, not measurements; the top-1 jackknife leverage in Table 7 flags the six cases where  $\sigma_{\text{jack}}$  rests on effectively one replica. For the four *measured* ones among them (Ceres, Juno, Psyche, Thisbe) we recomputed  $\sigma$  with a nonparametric bootstrap (resampling targets with replacement, warm-started from the converged fit), which is robust to single-replica dominance because a high-leverage target does not enter every resample. The bootstrap confirms the low-leverage pair (Ceres  $\sigma_{\text{boot}}/\sigma_{\text{jack}} = 0.94$ , Psyche 1.07; both stay  $\text{snr} > 3$ ) but shows the jackknife *underestimated*  $\sigma$  for the two highest-leverage cases: Juno (lev. 0.93,  $\sigma_{\text{boot}}/\sigma_{\text{jack}} = 1.9$ ,  $\text{snr}_{\text{boot}} = 2.4$ ) and Thisbe (lev. 0.84, ratio 4.0,  $\text{snr}_{\text{boot}} = 1.8$ ) both fall below the  $\text{snr} = 3$  identifiability threshold and are accordingly demoted from measured to bounds. This is the concrete payoff of the leverage diagnostic: two masses that a jackknife-only analysis would have published as measurements are, under the robust  $\sigma$ , only bounds — notably Thisbe, whose apparent  $z = -10.6$  against the literature collapses to a statistically unremarkable  $z = -2.8$  once its uncertainty is honestly sized.

Estimator injection–recovery. That the estimator itself is unbiased — separate from the force-model accuracy anchored by the calibrators — is verified by an injection–recovery test: a known perturber mass is injected into synthetic per-transit astrometry of a set of test bodies (controlled encounter geometry, realistic along-scan noise) and recovered by the same joint mass+orbit fit. Injecting  $2 \times 10^{19}$  kg into 24 targets at 2 mas per-transit noise recovers  $2.18 \times 10^{19}$  kg ( $z = +0.5$ ,  $\chi_{\text{red}}^2 = 0.88$ ), with the jackknife  $\sigma$  again exceeding the formal  $\sigma$  by the expected factor; the recovery is unbiased across injected masses and seeds. The test is kept as a permanent regression guard.

Beyond the sixteen. The engine is not restricted to the sixteen ephemeris perturbors: each perturber is integrated as a massive



**Table 7.** The sixteen perturber masses from the joint orbit+mass fit to Gaia FPR astrometry.  $N$  = test bodies;  $\sigma_{\text{tot}}$  = total (formal  $\oplus$  systematic) fractional uncertainty; ratio = fit / reference (literature for the four calibrators, DE441 seed otherwise);  $z$  uses  $\sigma_{\text{tot}}$ ; lev. = top-1 jackknife leverage (fraction of the jackknife variance from the single most influential replica;  $> 0.5$  means  $\sigma_{\text{jack}}$  has  $\sim 1$  effective degree of freedom and is not defensible as a  $\sigma$ ); status from the identifiability criterion (measured / bound / non-physical / unconstrained). For the four *measured* cases with lev.  $> 0.5$  (Ceres, Juno, Psyche, Thisbe) the quoted  $\sigma_{\text{tot}}$  and status use the leverage-robust bootstrap  $\sigma$  (text); this leaves Ceres and Psyche measured but demotes Juno and Thisbe to bounds.

| perturber        | $N$ | mass (kg)             | $\sigma_{\text{tot}}$ | $\chi^2_{\text{red}}$ | ratio | $z$  | lev. | status        |
|------------------|-----|-----------------------|-----------------------|-----------------------|-------|------|------|---------------|
| (1) Ceres        | 28  | $9.0 \times 10^{20}$  | 6 %                   | 0.99                  | 0.96  | -0.7 | 0.68 | measured      |
| (2) Pallas       | 8   | $2.5 \times 10^{20}$  | 7 %                   | 0.99                  | 1.20  | +2.2 | 0.27 | bound         |
| (3) Juno         | 24  | $2.0 \times 10^{19}$  | 42 %                  | 0.99                  | 0.70  | -1.0 | 0.93 | bound         |
| (4) Vesta        | 28  | $2.4 \times 10^{20}$  | 6 %                   | 0.97                  | 0.94  | -1.1 | 0.27 | measured      |
| (7) Iris         | 27  | $7.1 \times 10^{18}$  | 119 %                 | 0.98                  | 0.42  | -1.2 | 0.62 | bound         |
| (10) Hygiea      | 26  | $8.2 \times 10^{19}$  | 7 %                   | 0.98                  | 0.99  | -0.2 | 0.43 | measured      |
| (15) Eunomia     | 29  | $2.3 \times 10^{19}$  | 31 %                  | 0.98                  | 0.77  | -1.0 | 0.32 | measured      |
| (16) Psyche      | 26  | $2.3 \times 10^{19}$  | 9 %                   | 0.98                  | 0.95  | -0.6 | 0.64 | measured      |
| (31) Euphrosyne  | 26  | $2.0 \times 10^{19}$  | 45 %                  | 0.99                  | 1.26  | +0.5 | 0.23 | bound         |
| (52) Europa      | 28  | $2.3 \times 10^{19}$  | 17 %                  | 0.98                  | 0.58  | -4.3 | 0.43 | measured      |
| (65) Cybele      | 15  | $6.1 \times 10^{19}$  | 60 %                  | 0.93                  | 4.33  | +1.3 | 0.41 | bound         |
| (87) Sylvia      | 12  | $3.5 \times 10^{19}$  | 39 %                  | 0.99                  | 1.06  | +0.2 | 0.46 | bound         |
| (88) Thisbe      | 27  | $6.9 \times 10^{18}$  | 57 %                  | 0.98                  | 0.39  | -2.8 | 0.84 | bound         |
| (107) Camilla    | 2   | $1.7 \times 10^{20}$  | 67 %                  | 0.99                  | 8.07  | +1.3 | —    | unconstrained |
| (511) Davida     | 6   | $-1.1 \times 10^{20}$ | 50 %                  | 0.81                  | —     | -3.1 | 0.60 | non-physical  |
| (704) Interamnia | 29  | $2.0 \times 10^{19}$  | 40 %                  | 0.98                  | 0.47  | -2.8 | 0.31 | bound         |

rebound particle from its own orbital elements, so a seventeenth body outside sb441-n16 is added as a seventeenth massive particle with no double-counting (the ASSIST ASTEROIDS force is deliberately excluded). This was demonstrated on (19) Fortuna — orbit fixed from JPL Horizons, mass free — which fits with  $\chi^2_{\text{red}} = 0.977$  and  $M = 1.13 \times 10^{19} \pm 2.2 \times 10^{18}$  kg. The methodological gate is green: the engine generalises to bodies beyond the ephemeris set. Fortuna is not itself a new mass (it appears in Goffin 2014 and Fuentes-Muñoz et al. 2025), and its 20 %  $\sigma$  makes it a coarse determination consistent with the literature within the jackknife bar ( $z = +1.25$ ). The catalogue’s top ranked out-of-sixteen candidate, (9) Metis (37 useful encounters, Sect. 4.4), was run as a second out-of-sixteen case and fits with  $\chi^2_{\text{red}} = 0.981$  and  $M = 4.74 \times 10^{18} \pm 1.53 \times 10^{18}$  kg (55 targets, jackknife  $\sigma = 32$  %), a ratio of 0.73 against Fuentes-Muñoz et al. (2025) ( $6.48 \times 10^{18}$  kg), consistent within the jackknife bar ( $z = -1.14$ ). Fortuna and Metis thus scatter to either side of the literature (+30 % and -27 % in central value) but both remain consistent once the honest external  $\sigma$  is used — confirming the method generalises while underscoring that a single out-of-sixteen perturber with  $\sim 50$  Gaia-FPR targets yields only a coarse (20–30 %) determination, not a competitive mass. Neither is a new mass: both are already in Fuentes-Muñoz et al. (2025).

We tested whether the mean -4 % calibrator bias is an artefact of an incomplete perturber background by enlarging it from the sixteen ephemeris bodies to thirty-five — adding the twenty most massive asteroids of Fuentes-Muñoz et al. (2025) ( $\approx 1.6 \times 10^{20}$  kg spread across the belt), with masses from their Table 5 and orbits from JPL Horizons, using the same custom-perturber machinery. The calibrator masses shift by  $< 0.25$  % (Ceres -0.02 %, Vesta -0.23 %, Hygiea +0.18 %), three orders of magnitude below the deficit being probed; the systematic floor  $f_{\text{sys}}$  does not drop (4.16 %  $\rightarrow$  4.26 %). Background incompleteness is therefore *not* the origin of the -4 % bias — a far-field, dispersed perturber population averages to nearly zero deflection over the target ensemble. A scale bound confirms this: the thirty-five modelled bodies carry  $\approx 1.7 \times 10^{21}$  kg, leaving an unmodelled tail of only  $\sim 7 \times 10^{20}$  kg (against a total main-belt mass  $\approx 2.4 \times 10^{21}$  kg) spread over  $> 10^5$  objects

each below  $\sim 5 \times 10^{18}$  kg — individually an order of magnitude lighter and far more numerous, hence more dispersed, than the twenty just added. Since the 4  $\rightarrow$  35 enlargement (concentrating  $1.6 \times 10^{20}$  kg in the largest remaining bodies) moved the paired  $\Delta\text{mass}$  by  $< 0.25$  %, the residual tail — lighter per body and closer to isotropic — bounds to well below that gate and cannot produce a 4 % coherent deficit. The residual bias is therefore more plausibly attributable to per-target orbit imperfection over the short Gaia arc or unmodelled per-encounter astrometric systematics.

**Relation to other methods.** The estimator used here (a joint orbit+mass least-squares fit with an FOV-block covariance, an external jackknife  $\sigma$ , and an identifiability flag) is functionally a transparent reimplementation of established encounter-based mass determination, not a new methodology. Sitala & Granvik (2020) obtain full Bayesian posteriors by MCMC, which natively expose multi-modality and non-Gaussian tails that a single covariance cannot; Fuentes-Muñoz et al. (2025) propagate a prior to a posterior over the same Gaia FPR astrometry. Relative to those, our contribution in this section is not methodological sophistication but end-to-end demonstration that the open all-pairs catalogue feeds a mass pipeline whose every step (target selection, covariance model, jackknife, identifiability cut) is open and reproducible. Because Fuentes-Muñoz et al. (2025) fits the same FPR data, the cross-check below is a consistency test between two analyses of one dataset, not an independent confirmation.

## 6. Data availability

The primary product is the Kepler-refined candidate catalogue, `encounters_catalog_rebound_005au_b1fix.parquet` (80072774 rows), a typed Parquet file with a machine-readable provenance sidecar recording the MPCORB snapshot hash (full SHA-256), the frozen configuration, and the pipeline parameters. The characterised catalogue, `encounters_characterized_b1fix.parquet`, adds observability, solar elongation, magnitudes, measured/estimated

diameters (with a per-row provenance flag), a dynamical class, and a deflection-signal estimate for all 80 M rows (schema 710 Appendix A). A derived *hybrid* catalogue carrying per-row refinement\_method  $\in \{\text{kepler}, \text{nbody}\}$  and both Kepler and N-body geometry on the fragile subset — for science depending on sub-mAU distances — is regenerated from the same freeze. The catalogue and its input hashes (MPCORB snapshot and the JPL SBDB physical-data snapshot) are recorded in the sidecars; anything derived downstream must cite the catalogue SHA-256 and the code commit at its generation time. SHA-256 digests of the deposited files are listed with the archive record.

## 7. Conclusions

720 We have built and characterised a systematic catalogue of 80 072 774 real 3D close encounters ( $< 0.05$  AU) between numbered asteroids during the Gaia DR3 observation window. Unlike prior encounter-based work, which selects individual events by hand, the catalogue is exhaustive over the numbered population and — its distinguishing feature — comes with a *measured* completeness budget rather than an assumed one: the two-body refinement error (median  $12 \mu\text{AU}$ , p99  $2.5 \text{ mAU}$ ), the threshold-induced false-negative rate ( $0.70\%$ , symmetric and scatter-dominated, implying  $\sim 10^5$  censored encounters catalog-730 wide), and the orbital prefilter recall on the adverse high- $e$ /high- $i$  tail ( $76\%$ , with a zero-cost radial-overlap fix that recovers  $100\%$ ). A hybrid variant provides N-body-grade geometry on the dynamically fragile subset.

The catalogue is a working target-selection front-end for mass determination: a joint orbit+mass fit on Gaia FPR astrometry recovers the calibrators, determines the studied perturbers, and generalises to bodies beyond the ephemeris set, demonstrated end-to-end (Sect. 5). We are explicit about the limits: distances are candidates under frozen two-body assumptions except on the N-body-refined subset, the catalogue is not complete, and the large-large ( $D \gtrsim 100 \text{ km}$ ) approaches it contains all involve well-surveyed bodies with no unpublished large-body mass emerging — the mass layer is a demonstration of use rather than a source of competitive new masses. The contribution is the dataset itself: an all-pairs catalogue published in full with a quantified incompleteness budget, fed to a mass pipeline whose every step (FOV-block covariance, external jackknife  $\sigma$ , identifiability flag) is open and reproducible — a transparent reimplementation of established encounter-based mass determination 740 (Siltala & Granvik 2020; Fuentes-Muñoz et al. 2025), not a new methodology.

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## Appendix A: Catalogue column schema

Table A.1 documents the columns of the characterised catalogue (`encounters_characterized_full.parquet`), the widest of the three published products. The detection catalogue (`encounters_catalog_rebound_005au.parquet`) carries the first block of identity and geometry columns only (`number_1` through `rel_vel_m_s`); the hybrid catalogue (`encounters_catalog_hybrid_stageb.parquet`) adds a `refinement_method`  $\in \{\text{kepler}, \text{nbody}\}$  column and, for the N-body-refined rows, parallel `dist_au_nbody`, `jd_tdb_nbody`, `rel_vel_au_day_nbody` columns and their Kepler–N-body residuals. Bodies are ordered so that subscript 1 is the larger (perturber) body of the pair (Sect. 2.3).

**Table A.1.** Columns of the characterised close-encounter catalogue.

| column                 | type    | unit               | description                                                                                                 |
|------------------------|---------|--------------------|-------------------------------------------------------------------------------------------------------------|
| run_id                 | string  | —                  | Pipeline run identifier (ISO timestamp)                                                                     |
| number_1               | int32   | —                  | MPC number of body 1 (larger/perturber)                                                                     |
| number_2               | int32   | —                  | MPC number of body 2 (smaller/target)                                                                       |
| designation_1          | string  | —                  | MPC packed designation of body 1                                                                            |
| designation_2          | string  | —                  | MPC packed designation of body 2                                                                            |
| jd_tdb                 | float64 | d                  | Julian Date of closest approach (TDB)                                                                       |
| date_utc               | string  | —                  | Calendar date of closest approach (UTC)                                                                     |
| dist_au                | float64 | AU                 | Minimum 3D separation at closest approach                                                                   |
| dist_km                | float64 | km                 | Minimum 3D separation at closest approach                                                                   |
| rel_vel_au_day         | float64 | AU d <sup>-1</sup> | Relative velocity at encounter                                                                              |
| rel_vel_km_s           | float64 | km s <sup>-1</sup> | Relative velocity at encounter                                                                              |
| rel_vel_m_s            | float64 | m s <sup>-1</sup>  | Relative velocity at encounter                                                                              |
| H_1                    | float64 | mag                | Absolute magnitude of body 1 (MPCORB)                                                                       |
| H_2                    | float64 | mag                | Absolute magnitude of body 2 (MPCORB)                                                                       |
| diameter_1_km          | float64 | km                 | Diameter of body 1 (measured if available, else from $H$ )                                                  |
| diameter_2_km          | float64 | km                 | Diameter of body 2 (measured if available, else from $H$ )                                                  |
| diameter_source_1      | string  | —                  | Provenance: measured   albedo_measured   zone_albedo   default_albedo   unknown                             |
| diameter_source_2      | string  | —                  | Provenance of <code>diameter_2_km</code> (same categories)                                                  |
| deflection_dv_m_s      | float64 | m s <sup>-1</sup>  | Impulse-approximation velocity kick $2GM_1/(b v_{\text{rel}})$ on body 2; ranking metric, not a measurement |
| class_1                | string  | —                  | Dynamical class of body 1 (NEA/MBA/Trojan/Centaur/TNO/Other)                                                |
| class_2                | string  | —                  | Dynamical class of body 2                                                                                   |
| solar_elongation_deg   | float64 | deg                | Solar elongation of encounter midpoint (legacy)                                                             |
| solar_elongation_1_deg | float64 | deg                | Solar elongation of body 1 from Earth                                                                       |
| solar_elongation_2_deg | float64 | deg                | Solar elongation of body 2 from Earth                                                                       |
| app_mag_1              | float64 | mag                | Apparent $V$ magnitude of body 1 (HG model)                                                                 |
| app_mag_2              | float64 | mag                | Apparent $V$ magnitude of body 2 (HG model)                                                                 |
| gaia_observable_1      | bool    | —                  | Body 1 individually detectable by Gaia at encounter                                                         |
| gaia_observable_2      | bool    | —                  | Body 2 individually detectable by Gaia at encounter                                                         |
| gaia_observable        | bool    | —                  | Combined flag: <code>gaia_observable_1</code> OR <code>gaia_observable_2</code>                             |